STATS 8 Midterm 2 Practice

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Instructions:

For free response questions, write only in the provided space. If there is a final answer box, **please write your final numeric answer in the provided box**. Show your work or explain your reasoning in the provided space, so that if your final answer is incorrect we may still give partial credit. For all numeric answers, round to 4 decimal places (don't worry about significant figures, there will be slack in the grading for rounding). I recommend reading over the entire exam before starting!

The following expressions may be useful to you during the exam. They are provided so that you do not need to memorize certain burdensome formulas, but you are still responsible for recognizing their contexts.

$$\sum_{i=1}^{n} x_i P(X = x_i) \qquad \sum_{i=1}^{n} (x_i - E(X))^2 P(X = x_i) \qquad \frac{\sigma}{\sqrt{n}}$$

1. The National Health Interview Survey is a nationally representative household survey of the U.S. non-institutionalized population. In 2021, the survey showed that in adults aged 18 and over, 10.2% smoked cigarettes **only**, 3.2% used e-cigarettes **only**, and 1.3% both smoked cigarettes and used e-cigarettes.

   Let event A be the event that a randomly selected adult from this survey used e-cigarettes.
   Let event B be the event that a randomly selected adult from this survey used cigarettes.

   (a) What probability does 1.3% represent? Use the probability notation we learned in class.

   $P(A \textbf{ and } B)$

   (b) A student says: "Based on this information, $P(A) = 3.2\%$." Explain what is wrong with this statement.

   Solution: 3.2% is the probability that a randomly selected adult uses e-cigarettes *only*. This is $P(A \text{ and } B^c)$, the probability that they use e-cigarettes and *not* cigarettes.

(c) What is $P(A)$?
Work or explanation:

Solution: My preference is to draw this out with a Venn diagram, but we can use the law of total probability: $P(A) = P(A \text{ and } B^c) + P(A \text{ and } B) = 0.032 + 0.013 = 0.045$.

Final answer:

(d) What is $P(A \text{ or } B)$?
Work or explanation:

Solution: Because of how the probabilities are provided to us, this will just be their sum: $P(A \text{ or } B) = 0.102 + 0.032 + 0.013 = 0.147$.

Final answer:

(e) In plain English, what quantity does $P(A \mid B^c)$ represent in context?

Solution: $P(A \mid B^c)$ is the probability that a randomly selected person from this survey who does not use cigarettes, does use e-cigarettes.

(f) Are the events A and B independent? Why or why not?

Solution: They are independent if $P(A \text{ and } B) = P(A)P(B)$. We still need to compute $P(B)$, but we have the other two probabilities.
$P(B) = 0.102 + 0.013 = 0.115$, so we have:

$$P(A \text{ and } B) = 0.013,$$
$$P(A)P(B) = 0.045 \times 0.115 = 0.0052.$$

So, A and B are not independent.

2. A 2021 study published in the Journal of the American Medical Association (JAMA) looked at whether incentives for weight loss were effective for weight loss and maintenance in individuals with obesity. Participants were randomly assigned to one of four equal-sized groups (86 people per group): (1) usual care only (no intervention), (2) financial incentives (one intervention), (3) health messages by phone/email (one intervention), or (4) **both** financial incentives and health messages (two interventions).

(a) Let the random variable X = the number of interventions a randomly selected individual from this study received. Fill in the probability distribution function and cumulative distribution function table below.

x	0	1	2
$P(X = x)$	86/344	172/344	86/344
$P(X \leq x)$	86/344	258/344	344/344

(b) What is the expected value of the number of interventions a randomly selected person in this study received?

Work or explanation:

Solution: $E(X) = 0 \times \frac{86}{344} + 1 \times \frac{172}{344} + 2 \times \frac{86}{344} = 1$

Final answer:

(c) What is the variance of the number of interventions a randomly selected person in this study received?

Work or explanation:

Solution:

$$\begin{aligned}
\text{Var}(X) &= (0-1)^2 \times \frac{86}{344} + (1-1)^2 \times \frac{172}{344} + (2-1)^2 \times \frac{86}{344} \\
&= (-1)^2 \times \frac{86}{344} + 0^2 \times \frac{172}{344} + 1^2 \times \frac{86}{344} \\
&= \frac{86}{344} + \frac{86}{344} \\
&= \frac{172}{344} (= 0.50)
\end{aligned}$$

Final answer:

3. You can get a routine blood draw at one of two clinics. At Clinic A, the wait time is anywhere from 0 to 20 minutes, with all times equally likely. At Clinic B, the wait time is anywhere from 0 to 10 minutes, with all times equally likely. Clinic A is closer to home, but Clinic B usually has shorter waits, so you are trying to weigh your options.

Let X = the wait time at Clinic A, and let Y = the wait time at Clinic B.

(a) Which of the following describes the individual distributions of X and Y?

- ○ $X \sim \text{Normal}(0, 20)$ and $Y \sim \text{Normal}(0, 10)$
- ○ $X \sim \text{Bernoulli}(20)$ and $Y \sim \text{Bernoulli}(10)$
- √ $X \sim$ **Uniform**$(0, 20)$ **and** $Y \sim$ **Uniform**$(0, 10)$
- ○ $X \sim \text{Uniform}(20, 20)$ and $Y \sim \text{Uniform}(10, 10)$

For the remaining parts, assume X and Y have the distribution you selected in (a).

(b) What is $P(0 \leq X \leq 5)$?

Work or explanation:

Solution: This is the area of the rectangle with base width 5 and height $\frac{1}{20}$, so the area is $\frac{5}{20}$, or 0.25.

Final answer:

(c) If you go to Clinic A, what is the probability that you wait exactly 5 minutes?
Work or explanation:

Solution: The random variable is continuous, so the probability that it is exactly equal to 5 is 0.

Final answer:

(d) Assuming the wait times at the two clinics are completely unrelated (independent), what is the probability that the wait time would be at least 5 minutes at Clinic A **and** at least 5 minutes at Clinic B? Hint: Let A be the event that the wait time at Clinic A is at least 5 minutes, and let B be the event that the wait time at Clinic B is at least 5 minutes. Since the wait times are independent, the events A and B are independent.
Work or explanation:

Solution: $P(A) = 15 \times \frac{1}{20} = \frac{15}{20}$ (or 0.75) and $P(B) = 5 \times \frac{1}{10} = \frac{5}{10}$ (or 0.50). Since A and B are independent, $P(A \text{ and } B) = P(A)P(B) = \frac{15}{20} \times \frac{5}{10} = \frac{75}{200}$ or 0.375.

Final answer:

4. Suppose that the mean amount of time adults in a population of interest spend walking per day is 40 minutes, with a standard deviation of 20 minutes. A researcher plans to take a random sample of 100 adults and compute the sample mean walking time, $\bar{X}$.

   (a) Suppose the researcher repeats this process many times: each time, they randomly select 100 adults and compute the sample mean walking time. Around what value would the sample means be centered? Be sure to fill in the bubble entirely.

   ◯ 2 minutes
   ◯ 20 minutes
   √ **40 minutes**
   ◯ 100 minutes
   ◯ We cannot tell from the given information.

   (b) In one random sample of 100 adults, the sample mean walking time is 43 minutes. Which statement is most accurate? Be sure to fill in the bubble entirely.

   ◯ The next sample of 100 adults will also have a sample mean of 43 minutes.
   ◯ Our original hypothesis was incorrect, and the population mean walking time must actually be 43 minutes.
   √ **If we took a different random sample of 100 adults, we might get a different sample mean.**
   ◯ The value of $\bar{X}$ cannot vary because the population mean is given; it is 40.

   (c) What is the standard deviation of the sampling distribution of $\bar{X}$? Be sure to fill in the bubble entirely.

   ◯ About 0.2 minutes
   √ **About 2 minutes**
   ◯ About 20 minutes
   ◯ About 200 minutes
   ◯ We cannot tell from the given information.

   (d) Which of the following would be more surprising? Be sure to fill in the bubble entirely.

   ◯ One randomly selected adult walks 80 minutes per day.
   √ **A random sample of 100 adults has a sample mean walking time of 80 minutes.**

   (e) Which statement is most reasonable for samples of size 100? Be sure to fill in the bubble entirely.

   ◯ The distribution of individual walking times will become approximately normal.
   √ **The distribution of sample mean walking times will be approximately normal.**
   ◯ The sample mean walking time will always equal 40 minutes.
   ◯ The standard deviation of individual walking times will shrink from 20 minutes to 2 minutes.